

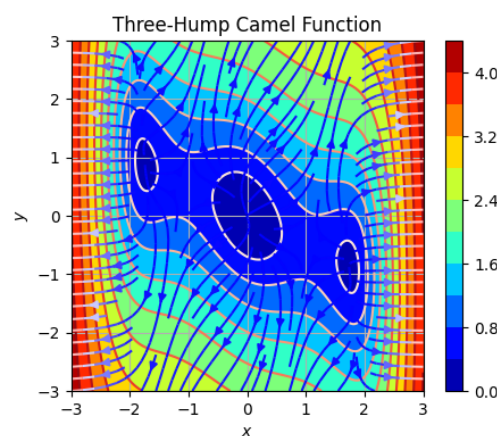
113A Introduction to Computer and Computer Science

Homework Assignment #9

Due: 11/25 12:00:00

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Problem #1: Three-Hump Camel Function (50%)

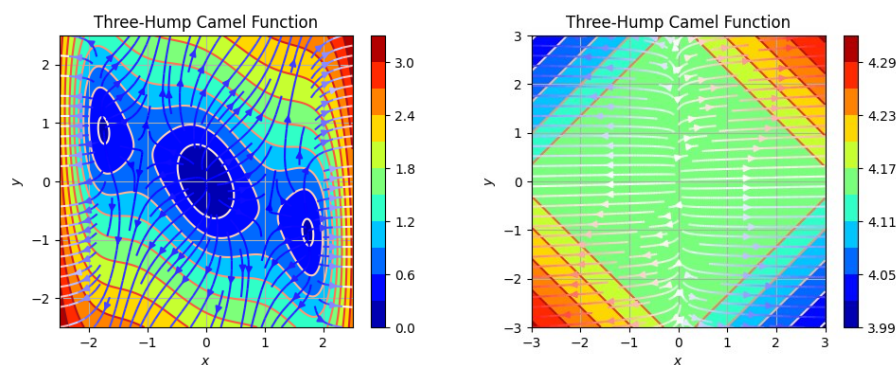


Please write a program to plot the contour map and draw the streamlines of its vector flow for the Three-Hump Camel function:

$$f(x, y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2$$

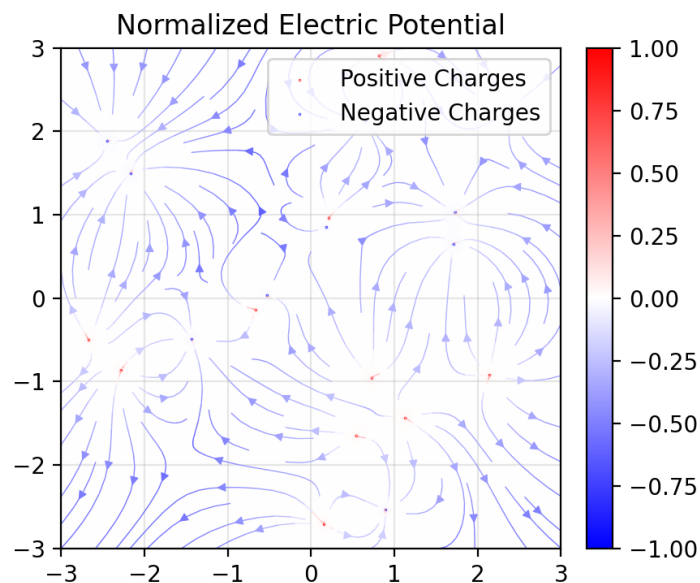
Please notice that,

1. The side length of your plot and the number of points along both axes should be adjustable for user. For example:



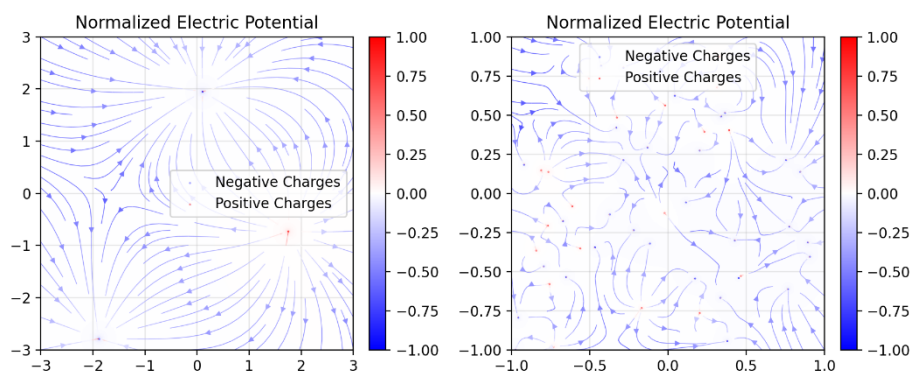
2. You don't need to follow the style of this example.

Problem #2: The electric field and electric potential of N random positive and negative charges (50%)



Please write a program to plot the normalized electric potential and draw the streamlines of its electric field for N random positive and negative charges in a 2-dimensional plane. Please notice that,

1. The side length of your plot, the number of points along both axes, and the number of random charges should be adjustable for user. For example:



2. Again, you don't need to follow the style of this example.

Please accomplish this homework with an organized code (e.g., with main script and function script). **Your main script file name must be 'main_hw9.py.'** Here is a template for your code structure:

```
113A_hw9_0123456789
├─ obj.py          # Module 1 for hw9
├─ func.py         # Module 2 for hw9
├─ :              # Other modules if necessary
└─ main_hw9.py     # Main script of hw9
```

FRIENDLY REMINDER

You don't need to follow this structure exactly, just keep your main script clean.

Here is a template for your main script.

```
import argparse
from solutions import solution1, solution2

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2024 NYCUDOPCS
Homework #9")
    parser.add_argument("--q", default=1, type=int, help="Which
question? (default: 1)")
    parser.add_argument("--l", default=3., type=float, help="Half side
length (default: 3.)")
    parser.add_argument("--N", default=501, type=int, help="Number of
points in 1D (default: 501)")
    parser.add_argument("--num", default=20, type=int, help="Number of
charges (default: 20)")
    parser.add_argument(
        "--save",
        default=False,
        action=argparse.BooleanOptionalAction,
        help="Save figure? (default: False)"
    )

    opt = parser.parse_args()
```

```
if opt.q == 1:
    print("Problem 1:")
    solution1(config=opt)

elif opt.q == 2:
    print("Problem 2")
    solution2(config=opt)

else:
    raise ValueError("The value of argument q should be 1 or 2.")
```

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
""  
Created on Wed Aug 7 01:23:45 2024  
  
@author: Xi Winnie, student ID  
  
@collaborators: William Lai, his student ID  
""
```

Please save your homework as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113A_hw9_ID.zip

For example,

113A_hw9_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.